BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: 92201

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. The value we get after rounding 1.47 is.

(A) 1.5
(B) 1.4
(C) 1.47
(D) 1.46

Q2. The slope of distance time curve only gives.

(A) Acceleration
(B) Speed
(C) Displacement
(D) Retardation

Q3. The value of g "$9.8ms^{-2}$" can be expressed in fts^{-2} as.

(A) $3.22fts^{-2}$
(B) $322fts^{-2}$
(C) $0.322fts^{-2}$
(D) $32.2fts^{-2}$

Q4. The mass of a body of weight 9.8N on the surface of earth will be.

(A) 9.8 Kg
(B) 0 Kg
(C) 1Kg
(D) 98Kg

Q5. How many times the centripetal force will increase if the mass of a body moving with uniform speed in a circle is doubled?

(A) Six times
(B) Two times
(C) Four times
(D) Eight times

Q6. S.I unit of torque is.

(A) Nm
(B) Nm^{-1}
(C) Nm^{2}
(D) Nm^{-2}

Q7. A body in equilibrium must not be.

(A) At rest
(B) Moving
(C) Rotating
(D) Accelerating

Q8. The magnitude of g on the moon's surface is about _____ of the value of g on the surface of Earth.

(A) 1/2
(B) 1/3
(C) 1/6
(D) 1/8

Q9. If the mass of a body becomes half and velocity becomes double the K.E will increase by a factor of.

(A) 2
(B) 3
(C) 4
(D) Will remains same

Q10. KWh is unit for.

(A) Energy
(B) Power
(C) Efficiency
(D) Pressure

Q11. A rock weighs 25.7N in air and 21.2N in water. What is the buoyant force of the water.

(A) 4.1N
(B) 3.5N
(C) 1.18N
(D) 0.84N

Q12. Absolute zero temperature on Celsius scale is.

(A) 273°C
(B) 100°C
(C) 0°C
(D) -273°C

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Find the number of seconds in a year and write the answer in standard form.

(ii) A ball is thrown upward with an initial speed of 5m/s. What will be its speed when it returns to starting point.

(iii) State Newton's First and second laws of motion.

(iv) Define rolling friction. Write methods of reducing friction.

(v) Can a small force ever exert a greater torque than a larger force? Explain.

(vi) Does Newton's law of Gravitation obay Newton's third law of motion? If Yes, then how?

(vii) Why for same height larger and smaller satellite must have same orbital speed?

(viii) A 60g bullet is fired from a gun with 3150J of kinetic energy. Find its velocity.

(ix) Explain how and why camels have adapted to allow them to walk more easily in desert conditions.

(x) The temperature of a normal human body is 37°C. Find this temperature on the Fahrenheit and kelvin scale.

(xi) Why burns caused by steam at 100°C on the skin are often more severe than burns caused by water at 100°C?

SECTION "C"

Maximum Marks: 21

Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [4] Derive the following equations of motion graphically. i) $S=vit + ½ at^2$ ii) $2as = vf^2 - vi^2$

(b) [3] A car of mass 1000kg is running on a circular motorway interchange near swabi with a velocity of $80ms^{-1}$, the radius of the circular motor way interchange is 800m. How much centripetal force is required.

QIV.

(a) [4] State and explain the conditions for complete equilibrium.

(b) [3] At which altitude above earth's surface would the gravitational acceleration be $4.9ms^{-2}$.

QV.

(a) [4] Prove that Kinetic energy of a body of mass m moving with velocity V is $K.E= ½ mv^2$.

(b) [3] An elevator weighing 5000N is raised to height of 15m in 10s, how much power is developed?

QVI.

(a) [4] How hydraulic lift works on the Pascal's principal? Explain.

(b) [3] How much heat is required to increase the temperature of 0.5kg of water from 10°C to 65°C. (Specific heat capacity of water is $4190JKg^{-1} K^{-1}$).